

Git Methods - Detailed Guide

Method 1: Command Line using HTTPS

This method teaches the core Git workflow using the terminal and an HTTPS remote URL.

Step 1: Check Git Installation

Command:

```
git --version
```

Explanation: Verifies that Git is installed.

Step 2: Initialize Repository

Command:

```
git init
```

Explanation: Creates a local Git repository.

Step 3: Check Status

Command:

```
git status
```

Explanation: Displays tracked and untracked files.

Step 4: Stage Files

Command:

```
git add .
```

Explanation: Stages all project files.

Step 5: Commit

Command:

```
git commit -m "Initial Commit"
```

Explanation: Creates the first snapshot.

Step 6: Rename Branch

Command:

`git branch -M main`

Explanation: Renames the default branch to main.

Step 7: Create GitHub Repository

Command:

`Create a new empty repository on GitHub.`

Explanation: Do not initialize it with README or .gitignore.

Step 8: Add Remote

Command:

*`git remote add origin
https://github.com/<username>/<repository>.git`*

Explanation: Links the local repository to GitHub.

Step 9: Verify Remote

Command:

`git remote -v`

Explanation: Shows fetch and push URLs.

Step 10: Push

Command:

`git push -u origin main`

Explanation: Uploads the repository to GitHub and sets the upstream branch.

Step 11: Daily Workflow

Command:

```
git status  
git add .  
git commit -m "Updated project"  
git push
```

Explanation: Typical workflow after making changes.

Step 12: Clone

Command:

```
git clone https://github.com/<username>/<repository>.git
```

Explanation: Downloads an existing repository.

Step 13: Pull

Command:

```
git pull origin main
```

Explanation: Downloads and merges the latest changes.

Step 14: History

Command:

```
git log --oneline
```

Explanation: Displays a compact commit history.

Method 2: VS Code Git Integration

VS Code provides a graphical interface for Git while executing Git commands behind the scenes.

Step 1

Open the project folder in VS Code.

Step 2

Open the Source Control panel.

Step 3

Click 'Initialize Repository' if the project is not yet a Git repository.

Step 4

Create a .gitignore file.

Step 5

Stage files using the '+' icon or 'Stage All Changes'.

Step 6

Enter a commit message (e.g., Initial Commit).

Step 7

Click the ✓ Commit button.

Step 8

Configure Git username and email if prompted:

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your_email@gmail.com"
```

Step 9

Click 'Publish Branch' or 'Publish to GitHub'.

Step 10

Sign in to GitHub if prompted.

Step 11

Choose Public or Private repository.

Step 12

After future edits: stage changes, commit, then click Push or Sync Changes.

Method 3: GitHub Desktop

Step 1

Install GitHub Desktop.

Step 2

Sign in with GitHub.

Step 3

File → Add Local Repository.

Step 4

Choose project folder.

Step 5

Review changed files.

Step 6

Enter commit message.

Step 7

Commit to main.

Step 8

Publish Repository.

Step 9

Choose Public or Private.

Step 10

Push future updates using Push Origin.

Common Git Commands

Command	Purpose
git init	Initialize repository
git status	Check status
git add .	Stage files
git commit -m "msg"	Commit changes
git remote -v	View remotes
git push	Push to GitHub
git pull	Download latest changes
git fetch	Fetch changes
git clone URL	Clone repository
git branch	List branches
git switch <branch>	Switch branch
git log --oneline	View commit history